Keywords Extraction With BERT

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***Abstract*—Keyword extraction is essential in modern Natural Language Processing (NLP) applications, serving as a gateway to more efficient information retrieval, document classification, and text summarization. In this project, we build upon KeyBERT—an unsupervised keyword extraction framework that uses BERT embeddings—by proposing two major extensions: refining embedding-based approaches and adapting the model for multilingual settings.**

First, we evaluate multiple embedding and generative models (e.g., Sentence Transformers, Flair, RoBERTa, vlt5 base, T5, and BART) under four distinct configurations, each balancing different degrees of precision, recall, coverage, and diversity. We compare these models on the Inspec dataset, measuring performance through ROUGE and cosine similarity metrics. Our experiments reveal that vlt5 base keywords typically achieves higher ROUGE scores, while BART excels in capturing semantic similarity across various extraction scenarios.

Second, we adapt KeyBERT for multilingual keyword extraction, focusing on French text using two datasets: WKC and Papyrus-f. We explore both multilingual encoders (e.g., paraphrase-multilingual-MiniLM-L12-v2, xlm-roberta-base) and French-specific models (e.g., Camembert), enhancing results through linguistic preprocessing steps such as lemmatization and named entity recognition. Additionally, we implement a semi-supervised learning strategy, splitting annotated data into training and test sets to fine-tune KeyBERT on domain-specific content.

Overall, our findings highlight the impact of parameter tuning, model selection, and language adaptation on keyword extraction performance. These extensions make KeyBERT more robust and versatile, paving the way for improved real-world applications in diverse linguistic contexts. All implementation details and code are available at https://github.com/malekiamir/KeyBERT-Extension.

I. Introduction

Keyword extraction is a fundamental task in Natural Language Processing (NLP) that involves identifying and extracting the most relevant terms or phrases from a given text. These keywords serve as concise representations of the document's main topics, enabling more efficient information retrieval, document classification, content summarization, and even sentiment analysis. As the amount of textual data continues to grow exponentially, the ability to efficiently identify key information from large datasets has become an increasingly important challenge.

In traditional keyword extraction methods, techniques such as frequency analysis or TF-IDF (Term Frequency-Inverse Document Frequency) have been widely used. However, these methods often fail to capture the contextual nuances of words, resulting in less accurate or irrelevant keyword selections. With the rise of deep learning and transformer-based models, more advanced approaches have emerged to address these limitations, offering better understanding of word meanings in context.

One such approach is KeyBERT, a state-of-the-art keyword extraction tool that leverages BERT (Bidirectional Encoder Representations from Transformers) to generate contextualized embeddings. By encoding the text into dense vectors that capture both the semantics and syntactic relationships of words, KeyBERT can identify keywords that are not only frequent but also most representative of the document's core topics. This method has proven to be more effective and reliable than previous techniques, especially when dealing with complex and varied text.

In this project, we explore the use of KeyBERT, a state-of-the-art keyword extraction tool based on transformer models. KeyBERT leverages BERT (Bidirectional Encoder Representations from Transformers) to generate contextualized embeddings, which are then used to identify the most salient keywords in a given document. This approach overcomes the limitations of traditional keyword extraction methods by considering the context in which words appear, making it more accurate and robust for modern applications.

Our project aims to enhance the performance of KeyBERT by incorporating multiple encoders and adapting the model for multilingual settings, specifically for French text. By doing so, we aim to improve the quality of keyword extraction and provide a more flexible tool for real-world applications in various languages.

II. Methodology

A. *Embeddings*

The methodology of this part of the project focuses on keyword extraction using transformer-based models and embedding techniques. We explored various pre-trained models for keyword extraction and conduct comparative analysis using evaluation metrics like ROUGE and Cosine Similarity. The project is based on two distinct approaches: (1) direct application of multiple embedding-based and generative models for keyword extraction and (2) a refined approach involving semi-supervised fine-tuning of KeyBERT. We utilized the **Inspec dataset** (*taln-ls2n/inspec*) containing scientific abstracts and their manually annotated keywords.

- **Baseline Implementation**
  We replicated the *"Keyword Extraction with BERT"* method and extended it by evaluating multiple embedding models and generative models:
  - **Embedding Models**: Sentence-Transformers(all-MiniLM-L6-v2), Flair, RoBERTa, SpaCy, Gensim, Universal Sentence Encoder (USE).
  - **Generative Models**: T5, BART.
  - **BERT base models**: bert keyword extractor, vlt5 base

  We extracted keywords using **KeyBERT** and evaluated them with:
  - **ROUGE scores**: Measuring keyword text overlap with ground truth.
  - **Cosine similarity**: Assessing semantic closeness between extracted and reference keywords.

  To improve keyword diversity, we also applied:
  - **Max Sum Similarity**: Selecting diverse yet relevant keywords.
  - **Maximal Marginal Relevance (MMR)**: Reducing redundancy in extracted keywords.

In this study, four different configurations and embedding conditions were explored to evaluate the impact of parameter selection on keyword extraction performance. Each configuration was designed to target a specific trade-off between precision, recall, diversity, and relevance.

- **Case One: Baseline Configuration**
  We use this configuration as a reference point with default settings that balance precision and recall. It ensures relevant keywords with moderate diversity, making it suitable for general use.
- **Case Two: High-Diversity Extraction**
  This setting enhances keyword variety by allowing more flexibility, considering both single and multi-word phrases, and relaxing stopword constraints. It also reduces repetition, promoting diverse outputs at the cost of some precision.
- **Case Three: Reliable and High Precision**
  This configuration favors precision, extracting fewer but highly relevant keywords by enforcing stricter rules and reducing randomness in sequence generation. It ensures consistency and is ideal when high confidence in keyword selection is essential.
- **Case Four: Extreme Coverage, Many Keywords)**
  In this configuration, we try to extract a large number of keywords, including diverse and multi-word phrases. It makes the search wider by allowing more stopwords and focusing on capturing as many relevant terms as possible. In generative models, parameters encourage increased randomness and exploration, ensuring that as many relevant keyphrases as possible are extracted. While this setting captures a wide range of keywords, it may introduce some noise due to its emphasis on inclusivity.
- **Semi-Supervised Learning**
  To enhance keyword extraction, we implemented a **semi-supervised learning** approach using the Inspec dataset. The data was split into:
  - **Training set (70%)**: Used to refine the KeyBERT model.
  - **Test set (30%)**: Used for final keyword extraction evaluation

  The refined model was evaluated using:
  - **ROUGE (ROUGE-1, ROUGE-2, ROUGE-L)**: Assessing precision and recall.
  - **Cosine similarity**: Comparing semantic representations of extracted vs. ground-truth keywords.

B. Multi-Language

In this section, we conducted experiments to enhance the performance of the KeyBERT model on languages other than English. We chose French as an example and two different datasets were used for this case:

- **WKC Dataset**
  - A relatively small keyword extraction dataset in French.
  - Contains 100 pieces of news documents and their corresponding ground-truth keywords.
- **Papyrus-f Dataset**
  - A larger and more diverse French dataset.
  - Includes academic paper titles and abstracts with labeled keywords.
  - Provides a more robust evaluation setting compared to WKC.

Several scenarios were tested to extract keywords:

- **Multilingual Models**
  These models were tested due to their broad language coverage, including French:
  - **paraphrase-multilingual-MiniLM-L12-v2 (baseline model)**
    A lightweight transformer-based model optimized for multilingual sentence embeddings. It is trained using a contrastive learning approach, enabling it to generate semantically meaningful vector representations across multiple languages.
  - **xlm-roberta-base**
    A multilingual transformer model based on RoBERTa, pre-trained using masked language modeling on 100 languages. It leverages a large-scale CommonCrawl dataset to learn robust cross-lingual representations.
- **French-Specific Models**
  These models were specifically designed for French and were expected to perform better:
  - **camembert-base**
    Unlike multilingual models, CamemBERT is trained solely on French text, making it highly optimized for the language. It is a transformer model based

on RoBERTa, pre-trained using masked language modeling on a large corpus of diverse French text.

- **French-Specific Enhancements**
  To further enhance keyword extraction in French, we implemented multiple preprocessing techniques, including:
  - *Lemmatization for Morphological Normalization*
    French has a rich morphological structure, where words appear in different forms due to conjugations, gender variations, and pluralization. We applied lemmatization using Stanza to convert words into their base forms before passing them to KeyBERT. (This helped reduce redundancy and improve recall in keyword extraction.)
  - *Named Entity Recognition for Important Terms*
    We observed that KeyBERT sometimes missed named entities (such as cities, organizations, and product names). To address this, we applied Named Entity Recognition using Stanza. (This significantly improved recall, as many missing key terms were added back to the final set.)
- **Fine-Tuned Multilingual Model on French Data**
  - Fine-tuning paraphrase-multilingual-MiniLM-L12-v2 on KWC and Papyrus-f
  - Fine-tuning xlm-roberta-base on KWC and Papyrus-f

  Using the given dataset, we first remove stopwords and then generate training pairs consisting of positive and negative examples. Next, we train the model for one epoch using cosine similarity loss.

For each model, the top five keywords were extracted as n-grams with lengths ranging from 1 to 3. The extracted keywords were then evaluated against ground-truth labels using ROUGE-L precision, recall, and F1 score.

III. Experiment and Results

A. *Performance of Embedding-Based Models*

For determining the impact and performance of our embedding models, we just analyzed the ROUGE and cosine similarity between keywords extracted by model and ground true values inside the dataset. We execute the models with different parameters in different cases. Table I illustrates different output metrics under the first case condition.

We have calculated the ROUGE-1, Rouge-2 and Rouge-L for all of the model but here just ROUGE-L is mentioned for all methods for the sake of making report briefly and short. In the first case we experience balanced approach, focusing on extracting a small number of concise and relevant keywords, it extracts only single words (Unigrams). We limit the number of extracted keywords to a moderate amount to prevent excessive or insufficient coverage and also, we use English common stopwords and there is no explicit diversity mechanism; all of these features have been set during fine tuning set of parameters of the model. For comparing the efficiency in equal condition, we have set equal parameters for the extractive and BERT based models and also same parameters for generative models have been set.

TABLE I
ROUGE SCORE AND COSINE SIMILARITY OF DIFFERENT EMBEDDING MODELS OF THE FIRST CASE (BASELINE CONFIGURATION)

Embedding Model	Rouge Score	Cosine Similarity
SentenceTransformer	rougeL: 0.2155	0.6394
Flair	rougeL: 0.1499	0.4551
SpaCy	rougeL: 0.1110	0.3226
gensim	rougeL: 0.0587	0.1457
Universal Sentence Encoder (USE)	rougeL: 0.1861	0.5229
RoBERTa	rougeL: 0.1134	0.322
BERT keyword extractor	rougeL: 0.2017	0.5636
vlt5 base keywords	**rouge1: 0.3860** **rouge2: 0.2245** **rougeL: 0.3065**	0.6390
T5	rougeL: 0.2143	0.6181
BART	rougeL: 0.2403	**0.6950**

TABLE II
ROUGE SCORE AND COSINE SIMILARITY OF DIFFERENT EMBEDDING MODELS OF THE SECOND CASE (HIGH DIVERSITY EXTRACTION)

Embedding Model	Rouge Score	Cosine Similarity
SentenceTransformer	rougeL: 0.2520	**0.7183**
Flair	rougeL: 0.1879	0.5855
SpaCy	rougeL: 0.1180	0.4005
gensim	rougeL: 0.1676	0.5454
Universal Sentence Encoder (USE)	rougeL: 0.1924	0.6180
RoBERTa	rougeL: 0.1261	0.4568
BERT keyword extractor	rougeL: 0.2384	0.6816
vlt5 base keywords	**rouge1: 0.3886** **rouge2: 0.2286** **rougeL: 0.3083**	0.6413
T5	rougeL: 0.2161	0.6325
BART	rougeL: 0.2428	**0.7146**

We can observe from Table I, the *vlt5 base keywords* embedding model outperforms other models in rouge score comparing with other models and *BART* model does a little bit better than other models like *SentenceTransformer*,*vlt5* and *T5*.

According to information which driven by using the embedding model in the second case in Table II, in this case, we extract both single-word and multi-word keyphrases that improve the coverage, we try to increase the number of extracted keywords which provides a more comprehensive summary. What is more, we allow the stop words to appear in the embedding to make it suitable for tasks where wider keyword extraction is valuable. In this case we can see that *vlt5 base keywords* did quite well compared to other embedding in rouge score and also, *SentenceTransformer* and *BART* did well with negligible difference.

In the third case, we tried to fine-tune the parameters of embeddings to have moderate and high precision embeddings from the text which are more relevant to the underlying text corpus. For this reason we strict the filtering limits extraction to unigrams, we try to reduce the number of extracted keywords by tuning top_n parameter ensuring that we keep the core concepts. We also apply English stop removal phase and not using MMR to disable diversity mechanisms, ensuring that only the highest-ranked terms are selected.

As observed from Table III, for the case that hyperparam-

TABLE III
ROUGE SCORE AND COSINE SIMILARITY OF DIFFERENT EMBEDDING MODELS OF THE THIRD CASE (CONSERVATIVE AND HIGH PRECISION)

Embedding Model	Rouge Score	Cosine Similarity
SentenceTransformer	rougeL: 0.1758	0.5703
Flair	rougeL: 0.1214	0.3839
SpaCy	rougeL: 0.0805	0.2458
gensim	rougeL: 0.0386	0.0957
Universal Sentence Encoder (USE)	rougeL: 0.1521	0.4547
RoBERTa	rougeL: 0.0881	0.2534
BERT keyword extractor	rougeL: 0.1631	0.4889
vlt5 base keywords	**rouge1: 0.2948** **rouge2: 0.1560** **rougeL: 0.2535**	0.5621
T5	rougeL: 0.1905	0.5343
BART	**rouge1: 0.2834** **rouge2: 0.1358** **rougeL: 0.2339**	**0.6331**

TABLE IV
ROUGE SCORE AND COSINE SIMILARITY OF DIFFERENT EMBEDDING MODELS OF THE FOURTH CASE (EXTREME COVERAGE)

Embedding Model	Rouge Score	Cosine Similarity
SentenceTransformers	rougeL: 0.1956	0.6807
Flair	rougeL: 0.1828	0.6449
SpaCy	rougeL: 0.1287	0.4555
gensim	rougeL: 0.1810	0.6132
Universal Sentence Encoder (USE)	rougeL: 0.1755	0.6579
RoBERTa	rougeL: 0.1368	0.5439
BERT keyword extractor	rougeL: 0.1945	0.6771
vlt5 base keywords	**rouge1: 0.3777** **rouge2: 0.2193** **rougeL: 0.2995**	0.6331
T5	rougeL: 0.2135	0.6669
BART	rougeL: 0.2425	**0.7299**

eters are tuned to focus on having conservative and high precision embeddings, the *BART* and *vlt5 base keywords* outperformed other methods in rouge score metric and again *BART* did well in cosine similarity metrics too since it was the closest result to the maximum similarity value of 1.

In the fourth case, we changed parameters to have extreme coverage of the extractor with many keywords, for this reason, the number of n-grams varies from 1 to 3 and we extracted 15 top keywords as our candidate keywords. It is obvious in Table IV that in this case *vlt5 base keywords* and *BART* embedding models again work well in extracting such kinds of keywords among different embeddings.

B. *Evaluating Multilingual and French-Specific Models*

To evaluate the impact of our modifications, we measured ROUGE-L Precision, Recall, and F1 Score after incrementally adding different techniques to the KeyBERT model. These improvements were applied to the paraphrase-multilingual-MiniLM-L12-v2 model using the WKC dataset. Table V summarizes the performance of each approach.

Table V shows the performance of the KeyBERT model as we incrementally added preprocessing techniques to improve keyword extraction for French text.

- Lemmatization had little impact on the results, as expected. Since it normalizes word forms (e.g., ”mangé” $\rightarrow$ ”manger”), it influenced similarity computation between extracted and reference keywords.
- Adding Named Entity Recognition (NER) slightly improved recall. This happened because some named entities were not explicitly present in the reference ground truth but were still relevant for keyword extraction.

TABLE V
ROUGE-L SCORES WITH INCREMENTAL IMPROVEMENTS

Model Version	**Precision**	**Recall**	**F1 Score**
Baseline (KeyBERT)	0.2868	0.2423	0.2566
+ Lemmatization	0.2757	0.2258	0.2425
+ Lemmatization + NER	0.2669	0.2934	0.2724

TABLE VI
ROUGE-L SCORES WITH INCREMENTAL IMPROVEMENTS

Model	**Dataset**	**Precision**	**Recall**	**F1 Score**
XLMR	WKC	0.2720	0.2402	0.2448
XLMR	Papyrus-f	0.1532	0.1896	0.1602
XLMR(finetuned)	Papyrus-f	0.1825	0.1982	0.1790
Camembert	WKC	0.2818	0.2410	0.2482
Camembert	Papyrus-f	0.1425	0.1729	0.1474

Since Camembert is a French-specific model, we expected it to perform better on a French dataset like Papyrus-f. However, XLM-R’s larger pretraining corpus may help it generalize better, especially if Papyrus-f includes complex linguistic structures that require broader multilingual representations. Another possibility is that Papyrus-f differs significantly from Camembert’s pretraining data (e.g., domain-specific language, terminology, or style). The fine-tuned XLM-R model achieves better results than the base XLM-R, confirming that fine-tuning helps the model adapt to the dataset’s specific vocabulary and domain. The increase in Recall (from 0.1896 to 0.1982) suggests that fine-tuning makes the model better at capturing more relevant keywords, even if some predictions are less precise. On the WKC dataset, Camembert performs slightly better than XLM-R, likely due to the nature of its pretraining data, which includes a substantial amount of French text from sources similar to news articles. This alignment allows Camembert to better capture the linguistic structures and keyword patterns found in WKC. However, the fine-tuned XLM-R model remains a strong contender, especially in multilingual contexts.

IV. DISCUSSION

Model performance is not solely dependent on the quality of the architecture but also on the characteristics of the training data. A model trained on a dataset with both diversity and domain similarity to the test data is more likely to perform well. Ensuring that training corpora cover a broad range of linguistic structures while closely matching the target domain is crucial for improving keyword extraction.